MILITARY SPECIFICATION

BALANCER, VEHICLE WHEEL, BUBBLE-TYPE, PORTABLE

This specification is approved for use by the Armament Research and Development Center (Tool and Equipment Engineering Directorate, Standardization and Specification Engineering Division), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a portable, bubble-type vehicle wheel balancer.

2 APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-60 - Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish complete balancers for the first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification . In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

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* 3.2 Design. The bubble type wheel balancer shall be new and shall meet the requirements of this specification. The balancer shall be designed to balance wheel and tire assemblies of passenger cars and trucks through 1.25 ton. The balancer shall include an operator controlled mechanism for engaging and disengaging the contact between the balancing head assembly and its mating pivot point. When tire and wheel assemblies are lifted from the balancing head, contact between the balancing head assembly and the mating pivot point shall automatically disengage thru a self activating mechanism without operator influence. The mechanism shall prevent pivot point and balancing head damage when the tire and wheel assemblies are being placed upon the balancer and removed from the balancer. The balancer shall require no external energy source.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the balancer industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The balancer shall be complete so that it can be used for the operations specified herein.

3.4.1 Fastening devices. All screws, pins, bolts, and fastening devices shall be provided with means for preventing loss of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.2 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching.

3.4.3 Welding, brazing and soldering. Welding, brazing and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.4 Threads. Threads shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.4.5 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with means for lubrication.

3.4.6 Interchangeability. All parts shall be manufactured to such standards as will permit replacement and adjustment without modification of parts or machine.

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3.5 Performance and product characteristics.

3.5.1 Bubble indicator. The balancer shall be equipped with a circular spirit level with a sensitivity of no less than 0.1 inch movement of bubble for zero degrees ten minute (10') angular displacement. The bubble balancer indicator shall be recessed in the balancing assembly head to prevent damage to the spirit level.

3.5.2 Centering cone and leveling plate. The balancer shall be equipped with a centering cone and leveling plate with accurately controlled surfaces. They shall accommodate all American passenger cars and trucks up to and including 1.25 ton having 2.00 through 5.00 inch inside diameter centering hole. If necessary, one machined adapter plate shall be furnished to fit wheel-to-hub bolt mounting holes of the 5, 6, and 8 hole patterns to accommodate vehicles 1.00 through 1.25 ton.

3.5.3 Dampening. The balancing head assembly of the balancer shall be equipped with a method or form of dampener to accommodate stabilization from oscillations or movement to completely stabilize within a period of 10 seconds from extreme amplitude or oscillating limits to stable level, using a balanced L78-15 (9.00 X 15) wheel-tire assembly.

* 3.5.4 Performance. The bubble balancer shall balance tire and wheel assembly L78-15 (9.00 X 15) and 6.75 X 16.5 inch wheel, Chrysler Part Number 4026506, with a mounted tire size of 9.50R X 16.5D, 8 ply. For information, the wheel has an eight hole lug pattern, a 6.5 inch diameter bolt circle, and a 4.78 diameter centering hole. The bubble balancer shall also accurately balance tire and wheel assemblies from the Commercial Utility Cargo Vehicle (CUCV) which is a General Motors truck, that is issued with two different sizes of wheels and tires identified as follows:

a. Wheel - 15 inch X 8 inch General Motors Part Number 14063307, six hole lug pattern, 5 1/2 inch diameter bolt circle with .66 inch offset.

   Tire - 10R15C, on/off road.

b. Wheel - 16 inch X 6.5 inch General Motors Part Number 14035374, eight hole lug pattern, 6.5 inch diameter bolt circle with .67 inch offset.

   Tire - LT235/85R-16E, on/off road.

The balance accuracy shall be such that once a tire and wheel assembly have been balanced, and the tire and wheel assembly and the balancer head marked for rotational position relative to one another, the tire and wheel assembly can be lifted off of the balancer head, turned 90 degrees relative to the balancer head, replaced on the head, the balancer shall indicate no more unbalanced condition than that which requires a .25 ounce or less wheel weight to return the bubble to the center of the level indicator circle. The same accuracy shall be repeatable when the tire and wheel assembly are removed, rotated, and replaced in the same manner at 180 degrees, 270 degrees, and 360 degrees of rotation.

3.5.5 Balance base. The balancer shall be provided with a base for floor mounting. The base may be fabricated of metal or high impact plastic. The base shall be provided with externally adjustable leveling device(s) to permit three-point leveling. The balancer shall be capable of being leveled using its own balancer head spirit level bubble. The base shall be designed to assure the balancer of meeting the performance requirements specified in 3.5.1, 3.5.3 and 3.5.4.

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3.6 Tools. The balancer shall be provided complete with all tools to perform its intended use, including passenger car and truck wheel weight removal and application tools.

3.7 Wheel weights. Unless otherwise specified (see 6.2.1), there shall be included with the balancer separate boxes of 50 each of 0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00, 2.50 and 3.00 ounce wheel weights and separate boxes of 25 each of the 3.50 and 4.00 ounce wheel weights for passenger cars. There shall also be included separate boxes of 25 each of 0.5, 1, 2, 3, 4, 5 and 6 ounce wheel weight for Chrysler wheel Part No. 4026506 and for trucks up to 1.25 ton.

3.8 Finish. The finish of all parts shall be clean, smooth, and free of burrs, sharp edges, and sharp corners except when required in the original design. Heavy wear surfaces shall be hardened stainless steel or hard chrome plated.

* 3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing is required and shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces to be painted shall be clean and free of all foreign matter detrimental to painting.

3.10 Identification marking. The balancer shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.11 Workmanship. Standards of workmanship shall be such as to assure equipment with characteristics found in the best standard commercial units. The balancer shall be free from all imperfections which could adversely affect appearance, function and serviceability.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination of one complete balancer and all tests under 4.5. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

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4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of "a" through "c" as follows. Failure of any unit to pass an examination, test, or inspection shall be cause for rejection.

a. Examination (see 4.4.2).

b. Performance test (see 4.5.1).

c. Packaging inspection (see 4.6).

* 4.4.1 Sampling. Sampling shall be used for lot acceptance in accordance with MIL-STD-105, Inspection Level II. The Acceptable Quality Level (AQL) for examination and test shall be 0.65 percent defective. Packaging shall be sampled in accordance with General Inspection Level I and an AQL set at 4.0.

* 4.4.2 Examination. Visually and manually examine each balancer to determine conformance with the requirements of 3.2 through 3.5.2, and 3.5.5 through 3.11. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the balancer and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.5 Tests. The balancer shall be positioned and leveled on a hard surface, initially set up in accordance with the manufacturer's instructions, and then subjected to the following tests to determine compliance with section 3 herein and the contract.

* 4.5.1 Performance. The bubble balancer shall balance the tire and wheel assemblies L78-15 (9.00 X 15) and the 6.75 X 16.5 wheel (Chrysler Part No. 4026506) with a mounted tire size of 9.50R X 16.5D, 8 ply. For information, the wheel is 8 hole, 6.5 inch bolt circle, and 4.78 diameter centering hole. The bubble balancer shall also accurately balance the tire and wheel assemblies identified as follows: Wheel - 15 inch X 8 inch General Motors Part Number 14063307, six hole lug pattern, 5 1/2 inch diameter bolt circle with .66 inch offset. Tire - 10R15C, on/off road; and Wheel - 16 inch X 6.5 inch General Motors Part Number 14035374, eight hole lug pattern, 6.5 inch diameter bolt circle with .67 inch offset. Tire - LT235/85R-16E, on/off road. After the tire and wheel assembly has been balanced, it shall be index marked in relationship to the centering cone and leveling plate. The tire and wheel assembly shall then be lifted from the centering cone and leveling plate and repositioned with respect to the index mark on the centering cone and leveling plate through 90 degrees, 180 degrees, 270 degrees and return to zero degrees. At each and all rotated positions it shall require no more than a 0.25 ounce wheel weight to return the bubble to the center of the bubble indicator (see 3.5.4). The balancing head engagement/disengagement function shall be observed for proper operation (see 3.2).

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4.5.2 Dampening. Install a previously balanced L78-15 (9.00 X 15) wheel and tire assembly on the balancer. Push down on the side wall of the tire until the balancer hub prevents further movement. Suddenly release the wheel so that the wheel oscillates about the hub pivot point. Measure the time required for the wheel to stop oscillating (i.e., bubble no longer crosses center of bulls-eye). The time shall not exceed 10 seconds (see 3.5.3).

4.5.3 Bubble sensitivity. The balancer head assembly shall be placed in the rest position. The spirit level bubble shall be checked to verify its centering in the bulls-eye. The balancer head assembly shall then be engaged with the pivot point and rotated through an angular displacement of 360 degrees, checking for bubble movement every 45 degrees. The resulting bubble movement shall be measured by any suitable means having an accuracy of at least 0.01 inches. This movement shall be less than 0.1 inches.

* 4.6 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of section 5.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or B, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Components of the balancer and hand tools shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.2 Preservative application. Immediately after cleaning and drying, apply a coat of type P-2 preservative to all exterior ferrous metal surfaces of the balancer and tools. Caster bearings shall be lubricated with type P-7 preservative oil. Preservatives specified and procedures for application shall conform to MIL-P-116.

5.1.1.3 Unit pack. The balancer shall be disassembled to effect savings in cubage and protect parts that are vulnerable to damage. Components which have been coated with type P-2 preservative compound shall be loosely covered or wrapped with barrier material conforming to MIL-B-121, grade A. The barrier shall be secured in place with tape conforming to PPP-T-60. The balancer shall be packed in a fiberboard box conforming to PPP-B-636, class weather-resistant. The components shall be blocked inside the unit container in a manner which shall restrict movement and protect parts against damage. The type of materials to be utilized and the method of application shall conform to MIL-P-116. The container closure shall be in accordance with the appendix to the box specification.

5.1.2 Level B. Preservation shall be the same as level A (see 5.1.1) except that type P-7 preservative oil shall be substituted for type P-2 preservative.

5.2 Packing. Packing shall be level A or B, as specified in the contract (see 6.2.1).

5.2.1 Level A packing. Each balancer shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The box shall be closed and strapped in accordance with the appendix to the applicable box specification.

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5.2.2 Level B packing. Packing shall be the same as level A (see 5.2.1) except the shipping containers shall conform to domestic type and class 1, respectively.

5.3 Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Industrial packaging (preservation, packing, and marking). When industrial packaging is specified in the contract or order (see 6.2.1), it shall conform to ASTM D3951-82.

6 NOTES

6.1 Intended use. This specification covers a portable, bubble-type vehicle wheel balancer for balancing original equipment wheels of passenger cars and trucks through 1.25 ton rated capacity.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. First article, when required (see 3.1).

c. Wheel weights, if different (see 3.7).

d. Painting, if different (see 3.9).

e. Identification marking, if different (see 3.10).

f. First article inspection, if different (see 4.3).

g. Selection of applicable level of protection A, B, or Industrial (see 5.1, 5.2 and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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